Quester Tangent Corporation

Denver Regional Transportation District (Denver RTD)
Contract N8US-30-F0003

Monitoring and Diagnostic System (MDS)

Car Monitoring Network Interoperability Test Procedures

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1 Interoperability Test

1.1 Scope

This document describes the test procedures for verifying interoperability of the Central Diagnostic Terminal (CDT) with other subsystems on the Eagle P3 car, in accordance with the Car Monitoring Network ICD (DDN-0432-0080) and individual subsystem ICDs.

1.2 Objective

The objective of the interoperability testing is to ensure that each subsystem meets the design specification. Each test is focused to ensure the subsystem has met the stated requirements. The tests are designed to mitigate risks for subsystems prior to vehicle integration testing, which follows delivery of equipment for installation on the Pilot Car.

1.3 Testing Overview

The testing will be performed between the CDT and each subsystem before the equipment is integrated into the vehicle. Unless otherwise agreed, this test will be performed by Quester Tangent at Quester Tangent's facilities with the assistance of the subsystem supplier.

The subsystem presented for testing shall exhibit its full car network functionality. If it is not practical to perform network testing with the complete physical subsystem, the subsystem supplier shall provide a means to simulate the behaviour of the complete physical subsystem as seen through its network interface to the CDT.

The subsystem supplier shall complete in-house testing of the Car Monitoring Network interface as part of the subsystem type testing before interoperability tests are performed.

Functional tests shall be performed for each distinct location of the supplier's subsystem within the car (e.g. HVAC#1, HVAC#2). Separate copies of the recording tables (see section 1.5.1) shall be used for each subsystem location. Participants in the testing should review options and decide on the method for testing multiple locations: either:

- One subsystem units is presented for testing, it is tested completely at the first location, then reconfigured to the second location and tests are repeated; or
- Multiple subsystem units are presented for testing, configured for each of the possible locations, and tests proceed in parallel.

Test equipment is described in section 3.

The tests verify the subsystem communications with the CDT under various operational conditions. The test framework creates normal and abnormal network conditions. These tests are mainly focused on Car Monitoring Network interface functional behaviour.

Network noise immunity testing as described in section 8 may be included as part of these tests to verify the subsystem in-house test results. The decision as to whether such tests are performed will be based on the nature of the subsystem presented for testing.

Note:	Testing must take place in an electronics laboratory with full static electricity control measures in place, and the testing must be conducted by personnel with training in electrostatic shock prevention techniques.

It is intended that these interoperability tests be performed in the sequence presented in this document.

1.4 Pre-Test Checklist

It is expected that the following activities will be completed several weeks before the interoperability tests described in the current document are performed:

1. Car monitoring network in-house test procedure review by Quester Tangent.
2. Subsystem supplier performed car monitoring network in-house type tests.
3. Car monitoring network in-house test report review by Quester Tangent.

The subsystem supplier is requested to use the Echelon application NodeUtil to perform the following queries on the network interface of the equipment under test, and to send the results to Quester Tangent for analysis before the interoperability tests.

1. Once only for each distinct type of equipment:
   - Neuron buffer configuration (NodeUtil Node Menu option ‘B’)
2. For each distinct car configuration (Car A, Car B) and location within the car (DOOR#1, DOOR#2, etc.):
   - Domain Table (NodeUtil Node Menu option ‘D’)
   - Application Configuration Structures (NodeIUtil Node Menu option ‘C’)
   - Node Address Table (NodeUtil Node Menu option ‘A’)
   - List Network Variables (NodeUtil Node Menu option ‘L’)
   - Network Variable Configuration Table (NodeUtil Node Menu option ‘N’)

1.5 Reporting

1.5.1 Recording Tables

Section 4 and later sections of this document constitute recording tables. These tables are used at the time of testing to record measured values and a pass or fail status for each test procedure. These hand-written sheets constitute a valid (draft) test report. Copies of these sheets will be incorporated into the final test report.

Section 4 records the identities of the personnel conducting and / or witnessing the tests, the part numbers, serial numbers and calibration dates of the test equipment used to conduct the tests and a list of support documentation used.

A copy of section 4 shall be completed for each piece of equipment that is tested using the procedures in this document. A new copy of section 4 shall be created if an individual procedure or group of procedures, is repeated or is conducted at a materially different time or location from the other procedures with the resulting data recording tables attached to the new front matter. A copy of section 4 shall also be created if a material change occurs in the test equipment or personnel.

Appendix A provides a place for comments regarding the test procedure document, test equipment, conduct of the tests, or the test outcome. Each test procedure recording table has a column labelled *Comment*. If the personnel performing a test consider a comment necessary or desirable, a comment number is placed in this column and the comment entered in a copy of the comments sheet. The structure of the Comments sheet supports the inclusion of as many comments, and sheets of comments, as required by the test activities.

1.5.2 Recorded Log Files

Quester Tangent will archive all log files captured during testing. Files will be available for review but will not be submitted as part of the formal report.

1.5.3 Formal Reporting

Quester Tangent will document and review the results of the interoperability test, based on the recording tables. The test report will be submitted to Rotem and the subsystem supplier for review.

2 References

This document shall be used in conjunction with the following documents and publications. If the referenced documents or publications are superseded by an approved revision, the most recent revision shall apply.

Note: The most current revision level of QT documents may be obtained by querying the QT document management system. The most current revision is assumed.

Table 1: Reference Documents

PUBLISHER	DOCUMENT NUMBER	TITLE
Quester Tangent	DDN-0432-0080	MDS Car Monitoring Network ICD
	DDN-0432-0085	Propulsion Car Monitoring Network ICD
	DDN-0432-0086	Friction Brakes Car Monitoring Network ICD
	DDN-0432-0087	Auxiliary Power Car Monitoring Network ICD
	DDN-0432-0088	HVAC Car Monitoring Network ICD
	DDN-0432-0090	Door System Car Monitoring Network ICD
	DDN-0432-0091	Communications Car Monitoring Network ICD
	DDN-0432-0093	Wayside Data Transfer ICD
	DDN-0432-0097	CCTV Car Monitoring Network ICD

3 Test Equipment

3.1 Basic Interoperability Test Framework

The basic interoperability test framework simulates the Car Monitoring Network on the Eagle P3 vehicle as shown in Figure 1.

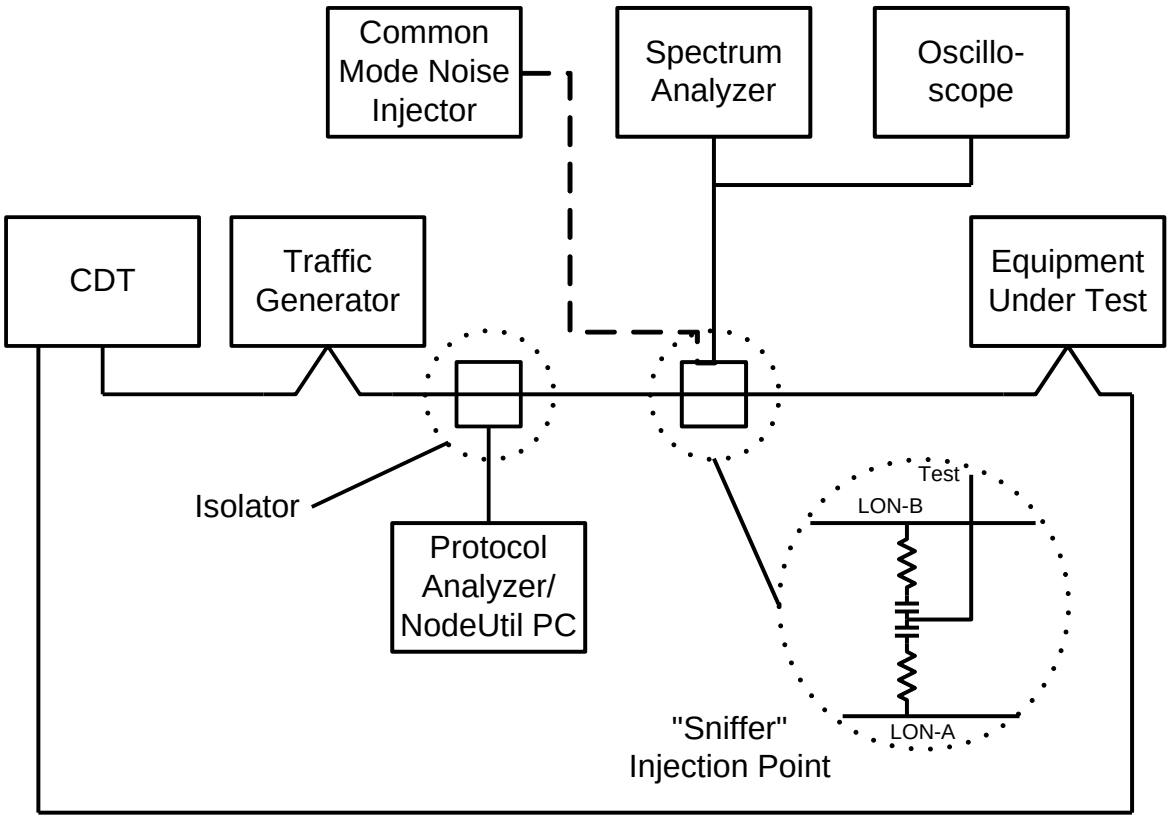

Figure 1: Interoperability Test Framework

3.1.1 Protocol Analyzer / NodeUtil PC

Protocol Analyzer is a software application that passively monitors LonWorks network traffic. NodeUtil is a software application that can be used to diagnose and configure LonWorks network devices. These applications run on a PC attached to the network via a standard COTS adapter.

3.1.2 Spectrum Analyzer and Oscilloscope

The spectrum analyzer and oscilloscope are used to monitor electrical noise on the LonWorks network.

3.1.3 Equipment Under Test (EUT)

The EUT is the third party subsystem being tested. The EUT interfaces with the CDT and test equipment through its subsystem vehicle interface. The subsystem supplier shall provide interface connector cables with bare wire connections for subsystem LVPS input power (37.5 VDC laboratory power will be provided) and the network interface. Any other EUT support requirements must either be provided by the subsystem supplier, or organized through discussions with Quester Tangent with sufficient lead-time to make the necessary arrangements.

The subsystem supplier shall provide all necessary means to verify EUT operations, manipulate EUT network status and parameter information and verify file transfer if required. Typically, the supplier's PTU software is used in such a support role; the supplier should expect to provide the PC that runs the software unless other arrangements are made with Quester Tangent.

The subsystem supplier shall be prepared to make (minor) changes to the EUT at the test location in the event of test failure, so that immediate re-testing shall be possible. In particular, it is essential that the supplier have the means to make updates to control and support software.

3.1.4 Central Diagnostic Terminal (CDT)

The CDT implements the car monitoring network interface and functionality as documented in DDN-0432-0080 (MDS Car Monitoring Network Interface Control Document, the "general ICD").

The CDT used in these tests can be prototype hardware and software; it does not require data logging, display or digital input and output functionality for interoperability testing.

CDT network interface shall conform to all timing requirements, network variable structures and file transfer protocols defined in the general and CDT ICDs.

CDT network functions include:

- Normal startup behaviour
- Transmit Subscribe network variables to all subsystems
- Receive StatusBits and Parameter network variables from all subsystems
- Detect loss of communication from any subsystem
- Transmit CDTParameter network variables to all subsystems
- Transmit CDTStatusBits network variables to all subsystems
- Support all incoming and outgoing file transfers described in the ICDs
- Support PTU interactions required for accessing other subsystems

For the purpose of testing, the file contents used in file transfer tests (SCI update, wayside transfers) need not represent final versions.

3.1.5 Traffic Generator

The Traffic Generator is custom test equipment consisting of a rack of multiple hardware modules connected to a common LonWorks network. The rack is powered from a COTS power supply.

The Traffic Generator simulates network nodes other than the CDT and EUT, and can optionally simulate a CDT if required. Each Traffic Generator module can simulate a single network node or multiple network nodes. The module behaviour is manually configured through the module's diagnostic serial port, attached to a PC running a terminal emulator.

The module configuration includes the following options:

- Simulate any single or multiple network nodes
- Generate normal network traffic
- Generate maximum network traffic
- Simulate intermittent network node behaviour
- Simulate network node power cycle behaviour
- Simulate asynchronous network events, e.g. Wayside transfer request
- Generate simulated SCI version information on request
- Simulate simulated SCI update file transfer behaviour
- Manually enter a time stamp and broadcast on command

Each module in the Traffic Generator is loaded with an identical software image. Its behaviour is configured through serial port interactions with the PC.

The proper behaviour of the Traffic Generator in simulating other network subsystems is validated through observations of network traffic using Protocol Analyzer.

3.1.6 Common Mode Noise Injector (CMNI)

The CMNI is an optional piece of equipment for the interoperability test framework. It is used to generate common mode noise on the network to artificially reduce the signal-to-noise ratio and increase the bit error

rate probability. By using the CMNI, it is possible to determine the susceptibility of the network nodes to electrical noise.

4 Car Monitoring Network Interoperability Test Procedure Equipment and Personnel Record

These tables form the front matter for the test procedures in this document. The individual test procedure results are recorded in tables associated with that procedure. When completed, the resulting document forms the basis of the test report.

4.1 Equipment Under Test Information

Subsystem – Configuration Sheet				
Description	Part Number	Revision	Software Name, Part Number, Revision, Version	Serial Numbers
Date:			Notes:	
Verified (QA):				

4.2 Inspection and Test Personnel Identification

INSPECTOR / TESTER FULL NAME	INITIALS	AFFILIATION	START DATE	COMPLETION DATE	LOCATION

4.3 Reference Documentation Record

REF	DOCUMENT TITLE	DOCUMENT NUMBER	REVISION
1			
2			
3			
4			
5			

4.4 Network Test Equipment Record

REF	INSTRUMENT TYPE	PART NUMBER AND REVISION	SOFTWARE PART NUMBER AND REVISION
1	CDT		
2	CDT PTU Software		
3	Traffic Generator		
4	Common Mode Noise Injector		
5	Protocol Analyzer software		
6	NodeUtil software		
7			
8			

4.5 Test Instrumentation Record

REF	INSTRUMENT TYPE	MODEL	SERIAL NUMBER	CALIBRATION DUE DATE
1	Spectrum Analyzer			
2	Oscilloscope			
3	Signal Generator			
4				
5				

4.6 EUT Location Configuration

EUT configured for car (A or B):_____ EUT configured for location within car (If applicable):_____

5 Normal Behaviour - All Subsystems

PROCEDURE SCOPE	Normal on vehicle operations. This procedure must be repeated for every locations (Car A, Car B and all possible EUT locations through the car if applicable)
PROCEDURE OBJECTIVE	Verify EUT normal behaviour.
REFERENCE DOCUMENTS	
TEST FACILITY	This procedure shall take place in an electronic test and maintenance facility with ESD protection.
TEST PERSONNEL	Inspection and test technician with ESD suppression training.

5.1 Network Configuration Tables

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect EUT and monitoring equipment according to the EUT location currently being tested.	N/A	N/A		
2.0	Configure EUT inputs for the location under test. Power on the EUT. Use the 'B' command on NodeUtil's Node menu to query EUT's Neuron buffer configuration. Record the settings: Type Count Size Receive transaction _____ _____ Transmit transaction _____ _____ App buffer in _____ _____ App buffer out _____ _____ Net buffer in _____ _____ Net buffer out _____ _____ App buff out priority _____ _____ Net buff out priority _____ _____	N/A	N/A		
3.0	Use the 'D' command on NodeUtil's Node menu to query EUT's domain table.	Domain table entry 0 has zero length, and subnet/node matches values in MDS Car Monitoring Network ICD section 5.1.			
4.0	Use the 'A' command on NodeUtil's Node menu to query EUT's address table.	Address table matches values in system-specific ICD for current location.			
5.0	Use the 'L' command on NodeUtil's Node menu to capture a listing of the EUT's network variables. Examine the file contents.	Network node and network variable self-documentation matches system-specific ICD.			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
6.0	Use the 'N' command on NodeUtil's Node menu to query EUT's network variable configuration table.	Network variable table matches values in system-specific ICD for current location.			
7.0	Use the 'C' command on NodeUtil's Node menu to query EUT's configuration table.	Program name and Location string match values in system-specific ICD for current location.			

5.2 Power On Tests

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment. Configure the EUT for a Car A location initially.	N/A	N/A		
2.0	Power off all nodes. Start logging network traffic with Protocol Analyzer.	No network traffic.			
3.0	Power on the EUT only. Record the network traffic with Protocol Analyzer for approximately one (1) minute and save the log to file. Record the file name below. _____	Only network traffic is EUT StatusBits packets.			
		EUT StatusBits packets are addressed to the Car A CDT.			
		EUT StatusBits are sent at a 9 seconds transmission period (± 250 ms).			
4.0	Power off all nodes. Start logging network traffic with Protocol Analyzer. Power on the Traffic Generator to simulate all nodes in Car, other than the CDT. Power on the EUT. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	EUT sends StatusBits with a 9 seconds transmission period (± 250 ms).			
5.0	Power on the CDT. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	CDT sends Subscribe packet for StatusBits to EUT within 2 seconds of receiving the first StatusBits packet from the EUT.			
		EUT sends StatusBits (and StatusBits2 if used) within 2 seconds of receiving a Subscribe packet from the CDT.			
		EUT sends StatusBits with a 2 seconds transmission period (± 250 ms).			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
6.0	Power off all nodes. Start logging network traffic with Protocol Analyzer. Power on the Traffic Generator to simulate all nodes in Car. Power on the CDT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize. Power on the EUT. Record the network traffic with Protocol Analyzer for up to one (1)minute and save the log to file. Record the file name below. _____	CDT sends Subscribe packet for StatusBits to EUT within 2 seconds of receiving the first StatusBits packet from the EUT.			
		EUT sends StatusBits (and StatusBits2 if used) within 2 seconds of receiving a Subscribe packet from the CDT.			
		EUT sends StatusBits with a 2 seconds transmission period (± 250 ms).			

5.3 Power Cycle Tests

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to the current testing location.	N/A	N/A		
2.0	Power off all nodes. Start logging network traffic with Protocol Analyzer.	No network traffic.			
4.0	Power off all nodes. Start logging network traffic with Protocol Analyzer. Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize. Power off the CDT for 10 seconds and power it on again. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	EUT detects loss of communications with CDT after 4 of its StatusBits packets are not acknowledged by the CDT.			
		EUT stops all network traffic and reverts StatusBits to a transmission period to 9 seconds (± 250 ms).			
		CDT sends Subscribe packet to EUT within 8 seconds of receiving a StatusBits packet from the EUT, after the CDT has finished startup.			
		EUT sends StatusBits with a 2 seconds transmission period (± 250 ms) after power cycle.			
5.0	Power off all nodes. Start logging network traffic with Protocol Analyzer. Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize. Power cycle a Traffic Generator node. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	EUT sends StatusBits with a 2 seconds transmission period (± 250 ms) throughout the power cycle.			

5.4 Time Synchronization Tests (All Except Communications Subsystem)

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to the current testing location.	N/A	N/A		
2.0	Power on the EUT. Set internal time to incorrect value if function exists. Verify the EUT time (log inspection, PTU connection, etc.).	EUT time is not present time or invalid.			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
3.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	Timestamp network variable is broadcast in response to CDT command.			
4.0	Verify the EUT time (log inspection, PTU connection, etc.).	EUT time is synchronized with "Comms" time.			
5.0	Disconnect the EUT from the network for > 9 seconds.	CDT detects loss of communications with EUT.			
6.0	Set EUT internal time to incorrect value. Verify the EUT time (log inspection, PTU connection, etc.).	EUT time is not present time or invalid.			
7.0	Re-connect the EUT to the network.	CDT sends Subscribe packet to EUT.			
	Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	Timestamp network variable is broadcast in response to CDT command.			
8.0	Verify the EUT time (log inspection, PTU connection, etc.).	EUT time is synchronized with "Comms" time.			
9.0	Disable the timestamp broadcast feature in the Traffic Generator.	CDT sends Subscribe packet to EUT.			
	Disconnect the EUT from the network for > 9 seconds. Set EUT internal time to incorrect value. Re-connect the EUT to the network.	CDT sends Command to Comms system, requesting Timestamp broadcast.			
	Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	After a delay (waiting for Comms broadcast), CDT broadcasts its own Timestamp network variable.			
10.0	Verify the EUT time (log inspection, PTU connection, etc.).	EUT time is synchronized with CDT time.			

5.5 Network Variable Update Tests

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to current testing location..	N/A	N/A		

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Access the CDT real time database monitoring functions to observe change of state for EUT StatusBits.	N/A	N/A		
4.0	Select an EUT monitored status data item from the StatusBits network variable to toggle. Record the StatusBits index and item name below. _____	N/A	N/A		
5.0	Trigger the selected status item to change from zero (0) to one (1). Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	Selected bit in StatusBits network variable state change observed with Protocol Analyzer.			
		Status item change of state observed on CDT.			
		CDT displayed change of state < 3 seconds after EUT state change.			
6.0	Trigger the selected status item to change from one (1) to zero (0). Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	Selected bit in StatusBits network variable state change observed with Protocol Analyzer.			
		Status item change of state observed on CDT.			
		CDT displayed change of state < 3 seconds after EUT state change.			
7.0	Select an EUT Parameter to observe. Record the Parameter index, name and CDT subscribe rate below. _____	N/A	N/A		
8.0	Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	The Parameter is sent by the EUT at the CDT subscription transmission period (± 250 ms).			
9.0	Set the EUT Parameter to its minimum value. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	EUT Parameter network variable with corresponding index number contains raw minimum value.			
		Minimum Parameter value observed on CDT.			
		CDT displayed change of state within 2 Subscribe transmission periods after Parameter value set on EUT.			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
10.0	Set the EUT Parameter to its maximum value. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	EUT Parameter network variable with corresponding index number contains raw maximum value.			
		Maximum Parameter value observed on CDT.			
		CDT displayed change of state within 2 Subscribe transmission periods after Parameter value set on EUT.			
11.0	Send a Subscribe update rate of 0 to the EUT Parameter being observed.	EUT stops sending updated value of selected Parameter.			
12.0	Re-send a Subscribe update rate of 0 to the EUT Parameter being observed.	EUT sends the selected Parameter value a single time to the CDT.			
13.0	Select a Parameter with a minimum update period in the CDT's "Primary Logging" configuration and a maximum update period in the "Alternate Logging". Record the Parameter name and update rates below. _____	N/A	N/A		
14.0	Change the logging option on the CDT from "Primary Logging" to "Alternate Logging". Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	CDT sends Subscribe packet to EUT with new update rates for all status bits and parameters.			
		EUT sends the Parameter selected in step 13.0 at the update period specified in the "Alternate Logging" after the Subscribe packet has been received from the CDT.			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
15.0	Change the logging option on the CDT back from “Alternate Logging” to “Primary Logging”. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends the Parameter selected in step 13.0 at the update period specified in the “Primary Logging” after the Subscribe packet has been received from the CDT.			
16.0	Select an EUT status data item from the StatusBits network variable. This item should correspond to a fault condition that is capable of being set and cleared rapidly. This status data item may be the same or a different one from the item tested in Step 4. Record the StatusBits index and item name (description) below. Index:_____ Name: _____	N/A	N/A		
17.0	Trigger the fault condition corresponding to the selected EUT data item and clear it again between one transmission of the StatusBits network variable and the next. The goal is to ensure that momentary faults are guaranteed to be transmitted to the CDT. Record the network traffic with Protocol Analyzer and save the log to file. Record the file name below. _____	Selected bit in StatusBits network variable transmitted as value 1 observed with Protocol Analyzer.			
		CDT displays (and logs) Status item state changes.			

5.6 Software Version Query Tests

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the current car configuration. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Access the CDT software version query functions and query the EUT. Record the results in the following table.	N/A	N/A		
4.0	Monitor the network traffic with Protocol Analyzer. Access the CDT software version query functions and query a Traffic Generator node. Monitor network traffic from the EUT. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	Software query does not affect periodic updates of EUT StatusBits and Parameter variables.			

SCI #	SCI TYPE		SCI NAME		SCI VERSION		PASS/FAIL	INTIAL	COMMENT
	EXPECTED	OBSERVED	EXPECTED	OBSERVED	EXPECTED	OBSERVED			

Note: The EUT subsystem supplier must provide the expected values in the above table prior to conducting the test.

5.7 Software Update Tests

This test procedure requires the EUT supplier to provide a valid software image that can be loaded onto the EUT. The image(s) must be distinguishable from the original software loaded on the EUT, e.g. by inspecting software version information. Quester Tangent will supply an invalid SCI file.

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to current testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Access the CDT–PTU software update functions and select the valid software image file for updating the EUT software. Record the file name below. _____	CDT–PTU is able to read software image information file without error.			
		Software image information file matches expected information.			
4.0	Update the EUT software with the valid image selected. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	CDT sends SCIUpdReq network variable to EUT with software image file information and "Initiate".			
		EUT sends SCIUpdResp network variable to CDT with "Initiate OK".			
		CDT and EUT exchange FileReq and FileStat variables before FTP begins.			
		File transfer as per "Echelon File Transfer Protocol". Data size is 32 bytes.			
		CDT and EUT exchange FileReq and FileStat variables after FTP ends.			
		CDT sends SCIUpdReq network variable to EUT with "Install" or "Install Reboot" as appropriate for the subsystem.			
		EUT sends SCIUpdResp network variable to CDT with "Install OK".			
		CDT reports software version information that matches with the software image update.			
5.0	Determine the installed software version from the EUT directly.	EUT software version information matches with the software image update.			
6.0	Restore the original software on the EUT.		N/A		

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
7.0	Access the CDT–PTU software update functions and attempt to update the EUT software with an invalid image file, e.g. an invalid checksum. Record the file name below. _____	CDT–PTU is able to read software image information file without error.			
		Software image information file matches expected information.			
8.0	Update the EUT software with the invalid image selected. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	CDT sends invalid image to EUT successfully up to the exchange of FileReq and FileStat network variables as tested in seq# 4.0.			
		CDT sends SCIUpdReq network variable to EUT with "Install" or "Install Reboot" as appropriate for the subsystem.			
		EUT sends SCIUpdResp network variable to CDT with "Initiate Refused".			
		CDT reports that EUT recognized the software image as invalid.			
		EUT software version information is unchanged.			
9.0	If possible, put the EUT into a state in which it is unable to handle a software update. Access the CDT–PTU software update functions and attempt to update the EUT software with the valid software image file.	CDT reports EUT refused software update.			
10.0	Monitor the network traffic with Protocol Analyzer. Access the CDT–PTU software update functions and command a (simulated) software update to a Traffic Generator node. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	EUT suspends network traffic during software image transfer.			
		EUT does not report loss of communications with CDT.			
11.0	Access the CDT–PTU software update functions and select the valid software image file for updating the EUT software. Command an update of the EUT software with the valid image selected. While the SCI file transfer is in progress, break the network connection between the CDT and the EUT.	CDT reports failure of file transfer.			
		EUT reports failure of file transfer.			
		CDT resumes normal operations.			
		EUT resumes normal operations.			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
12.0	Restore the network connection between the CDT and the EUT. Access the CDT–PTU software update functions and select the valid software image file for updating the EUT software. Update the EUT software with the valid image selected.	CDT updates the EUT software and EUT software version information matches with the software image update.			

5.8 File Transfer Tests

The following test verifies that when the EUT does not participate in a file transfer, it stops network transmissions during a log transfer and continues network transmissions during a wayside transfer. EUT does not report a loss of communications with the CDT as a result of either file transfer.

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to current testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car configuration. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Monitor the network traffic with Protocol Analyzer. Access the CDT–PTU log transfer functions and transfer a log file from a Traffic Generator node to the PTU.	EUT suspends network traffic during file transfer.			
		EUT does not report loss of communications with CDT.			
4.0	Generate a wayside file transfer using a Traffic Generator node.	EUT continues normal network activities during wayside file transfer.			

5.9 CDTStatusBits and CDTParameter Transfer

The following test verifies that the EUT correctly receives and processes transmissions from the CDT. The details of which CDT status bits and parameters are required by the EUT are documented in the subsystem-specific ICD. The execution of steps below will be tailored to the documented EUT data requirements.

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to current testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Observe the EUT's reporting of status bits received from the CDT. By appropriately stimulating the CDT, vary the status values transmitted.	EUT reports CDT status bits consistent with outputs of the CDT.			
4.0	Observe the EUT's reporting of parameter values received from the CDT. By appropriately stimulating the CDT, vary the parameter values transmitted.	EUT reports CDT Parameter values consistent with outputs of the CDT.			

6 Normal Behaviour - Subsystem Specific

PROCEDURE SCOPE	Normal on vehicle operations.
PROCEDURE OBJECTIVE	Verify EUT normal behaviour.
REFERENCE DOCUMENTS	
TEST FACILITY	This procedure shall take place in an electronic test and maintenance facility with ESD protection.
TEST PERSONNEL	Inspection and test technician with ESD suppression training.

6.1 Propulsion Subsystem - File Transfer Tests

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to current testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Cause the EUT to generate a log file.	N/A	N/A		
4.0	Connect the PTU to the CDT. Access the CDT–PTU log transfer functions. Verify the EUT reports files are available for transfer.	CDT–PTU reports that the log file is present in the EUT.			
5.0	Access the CDT–PTU log transfer functions and command a log file transfer from the EUT to the PTU. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	CDT and EUT exchange FileReq and FileStat variables before file transfer begins.			
		File transfer as per "Echelon File Transfer Protocol". Data size is 32 bytes.			
		CDT and EUT exchange FileReq and FileStat variables after file transfer ends.			
6.0	Re-connect the PTU to the EUT. Use the PTU software provided by EUT supplier to retrieve the logged data from the EUT. Compare the log files transferred to the CDT and accessed directly on the EUT.	PTU and EUT files have identical data content (formatting may differ).			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
7.0	Access the CDT–PTU log transfer functions and command a log file transfer from the EUT to the PTU. While the file transfer is in progress, break the network connection between the CDT and the EUT.	CDT reports failure of file transfer.			
		EUT reports failure of file transfer.			
		CDT resumes normal operations.			
		EUT resumes normal operations.			
8.0	Connect the PTU to the CDT. Access the CDT–PTU log transfer functions and select the function to clear the EUT log.	EUT reports log file is cleared.			

6.2 Friction Brakes Subsystem - File Transfer Tests

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Cause the EUT to generate a log file.	N/A	N/A		
4.0	Connect the PTU to the CDT. Access the CDT–PTU log transfer functions. Verify the EUT reports files are available for transfer.	CDT–PTU reports that the log file is present in the EUT.			
5.0	Access the CDT–PTU log transfer functions and command a log file transfer from the EUT to the PTU. Record the network traffic with Protocol Analyzer for up to one (1) minute and save the log to file. Record the file name below. _____	CDT and EUT exchange FileReq and FileStat variables before file transfer begins.			
		File transfer as per "Echelon File Transfer Protocol". Data size is 32 bytes.			
		CDT and EUT exchange FileReq and FileStat variables after file transfer ends.			
6.0	Re-connect the PTU to the EUT. Use the PTU software provided by EUT supplier to retrieve the logged data from the EUT. Compare the log files transferred to the CDT and accessed directly on the EUT.	PTU and EUT files have identical data content (formatting may differ).			
7.0	Access the CDT–PTU log transfer functions and command a log file transfer from the EUT to the PTU. While the file transfer is in progress, break the network connection between the CDT and the EUT.	CDT reports failure of file transfer.			
		EUT reports failure of file transfer.			
		CDT resumes normal operations.			
		EUT resumes normal operations.			
8.0	Restore the network connection between the CDT and the EUT. Access the CDT–PTU log transfer functions and transfer a log file from the EUT to the PTU. Re-connect the PTU to the EUT. Use the PTU software provided by EUT supplier to retrieve the logged data from the EUT. Compare the log files transferred to the CDT and accessed directly on the EUT.	PTU and EUT files have identical data content (formatting may differ).			

6.3 Communications Subsystem - Time Synchronization Tests

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect Traffic Generator, EUT and monitoring equipment according to testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Power on the CDT disconnected from the network. Set the internal time on the CDT to an incorrect value.	N/A	N/A		
4.0	Monitor network traffic with Protocol Analyzer. Connect the CDT to the network.	Comms System broadcasts a TimeStamp network variable in response to CDTCommand following start of communication.			
		CDT time synchronized with Comms time.			
5.0	Set the internal time on a Traffic Generator node to an incorrect value.	N/A	N/A		
6.0	Power cycle the Traffic Generator node with the incorrect time.	Comms System broadcasts a TimeStamp network variable in response to CDTCommand following node power cycle.			
7.0	Access Traffic Generator node time.	Traffic Generator node time synchronized with Comms time.			
8.0	Monitor network traffic with Protocol Analyzer for one or two hours.	Comms System broadcasts a TimeStamp network variable on an hourly basis.			

6.4 Communications Subsystem - Wayside File Transfer Tests

Note: Communications Subsystem shall be capable of file capture for verification of file transfers and simulation of any Wayside processing functions required to support the following test.

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Power on the EUT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	If possible, place network in a state to prevent the creation of new CDT log entries. Clear CDT logs.	N/A	N/A		
4.0	Start logging network traffic with Protocol Analyzer. Trigger EUT Recent Faults file transfer request to CDT. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends WaysideReq network variable to CDT with "Recent Faults" request.			
		CDT sends WaysideResp network variable to EUT.			
		WaysideResp network variable contains "No data" value.			
		EUT ends file transfer transaction when CDT reports no data available.			
5.0	Start logging network traffic with Protocol Analyzer. Trigger EUT Historical file transfer request to CDT for all available data. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends WaysideReq network variable to CDT with "Historical" request and "NULL" date.			
		CDT sends WaysideResp network variable to EUT.			
		WaysideResp network variable contains "No data" value.			
		EUT ends file transfer transaction when CDT reports no files available.			
6.0	Stimulate network to create CDT logs.	N/A	N/A		

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
7.0	Start logging network traffic with Protocol Analyzer. Trigger EUT Recent Faults file transfer request to CDT. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends WaysideReq network variable to CDT with "Recent Faults" request.			
		EUT continues normal network activities during file transfer.			
		Final WaysideReq variable contains the last CDT log entry timestamp.			
		WaysideResp network variable contains "No data" value.			
		EUT ends file transfer transaction when CDT reports no files available.			
8.0	Inspect file transferred to EUT and CDT log filtered for Recent Faults.	Files have identical data content (format may differ).			
9.0	Start logging network traffic with Protocol Analyzer. Trigger EUT Historical file transfer request to CDT for all available data. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends WaysideReq network variable to CDT with "Historical" request and "NULL" date.			
		EUT continues normal network activities during file transfer.			
		Final WaysideReq variable contains the last CDT log entry timestamp.			
		WaysideResp network variable contains "No data" value.			
		EUT ends file transfer transaction when CDT reports no files available.			
10.0	Inspect file transferred to EUT and CDT log filtered for Historical.	Files have identical data content.			
11.0	Stimulate network to create additional CDT logs.	N/A	N/A		

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
12.0	Start logging network traffic with Protocol Analyzer. Trigger EUT Historical file transfer request to CDT for data since last transfer. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends WaysideReq network variable to CDT with "Historical" request and date set to the last CDT log entry timestamp from previous "Historical" transfer.			
		EUT continues normal network activities during file transfer.			
		Final WaysideReq variable contains the last CDT log entry timestamp.			
		WaysideResp network variable contains "No data" value.			
		EUT ends file transfer transaction when CDT reports no files available.			
13.0	Inspect file transferred to EUT and CDT log filtered for Historical.	EUT file contains only the new CDT Log entries.			
14.0	Stimulate network to create CDT log entries.	N/A	N/A		
15.0	Trigger EUT Recent Faults file transfer request CDT. While the file transfer is in progress, break the network connection between the CDT and the EUT.	CDT reports failure of file transfer.			
		EUT reports failure of file transfer.			
		CDT resumes normal operations.			
		EUT resumes normal operations.			
16.0	Restore the network connection between the CDT and the EUT. Trigger EUT Recent Faults file transfer request to CDT. Inspect file transferred to EUT and CDT log filtered for Recent Faults.	Files have identical data content.			

7 Abnormal Conditions - Network Traffic Overload

PROCEDURE SCOPE	Network traffic overload conditions.
PROCEDURE OBJECTIVE	Verify EUT is robust under network traffic overload conditions.
REFERENCE DOCUMENTS	
TEST FACILITY	This procedure shall take place in an electronic test and maintenance facility with ESD protection.
TEST PERSONNEL	Inspection and test technician with ESD suppression training.

7.1 Network Traffic Overload

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to testing location.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Start logging network traffic with Protocol Analyzer. Set Traffic Generator to produce Network Traffic Overload condition. (Network Traffic Overload is defined as network traffic exceeding 75% when measured by the Protocol Analyzer software.) Power on the EUT. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	CDT sends Subscribe packet to EUT within 2 seconds of receiving the first StatusBits packet from the EUT.			
		EUT sends StatusBits within 2 seconds of receiving a Subscribe packet from the CDT.			
		EUT sends StatusBits with a 2 seconds transmission period (± 250 ms).			
		All EUT subscribed Parameter items are sent with the subscribed transmission period (± 250 ms).			
		The CDT reports no loss of communication with any subsystem.			
		The EUT reports no loss of communication with the CDT.			

7.2 Network Traffic Overload - Maximum Cable Length

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment according to testing location. EUT is connected to the test setup with no less than 50 meters of cable.	N/A	N/A		
2.0	Power on the Traffic Generator to simulate all other nodes in the car. Power on the CDT. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
3.0	Start logging network traffic with Protocol Analyzer. Set Traffic Generator to produce Network Traffic Overload condition. (Network Traffic Overload is defined as network traffic exceeding 75% when measured by the Protocol Analyzer software.) Power on the EUT. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	CDT sends Subscribe packet to EUT within 2 seconds of receiving the first StatusBits packet from the EUT.			
		EUT sends StatusBits within 2 seconds of receiving a Subscribe packet from the CDT.			
		EUT sends StatusBits with a 2 seconds transmission period (± 250 ms).			
		All EUT subscribed Parameter items are sent with the subscribed transmission period (± 250 ms).			
		The CDT reports no loss of communication with any subsystem.			
		The EUT reports no loss of communication with the CDT.			

8 EMI / EMC Compatibility

Execution of the tests in this section is optional. Quester Tangent will make the decision as to whether or not to conduct these tests, in consultation with the subsystem supplier. The decision will be based on review of subsystem design and testing, an assessment of potential risks during vehicle integration, and observations of the equipment presented for testing.

8.1 EMI / EMC Compatibility – Signal to Noise Ratio – Differential Noise Emissions

PROCEDURE SCOPE	EMI / EMC compatibility – Signal to Noise Ratio – Differential Noise Emissions.
PROCEDURE OBJECTIVE	Verify EUT transmit level is within requirements and that it does not emit excessive levels of differential noise onto the network conductors.
REFERENCE DOCUMENTS	EIA –709.3 Free Topology Twisted Pair Channel Specification (as referenced by IEEE 1473) IEC 62236-3-2 Railway applications – Electromagnetic compatibility Section 7, Table 5.
TEST REQUIREMENT	Transmit Signal Level > 0.425 Vrms, Differential Noise Level < 60 mVrms. These values represents > 20 dB Signal to Noise ratio at the LonWorks worst case specified transmit signal amplitude. The noise voltage level corresponds closely with the 99 dBµV (quasi peak) emission levels in the 150 kHz to 500 kHz spectrum as referenced in IEC 62236-3-2.
TEST FACILITY	This procedure shall take place in an electronic test and maintenance facility with ESD protection.
TEST PERSONNEL	Inspection and test technician with ESD suppression training.

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect CDT, Traffic Generator, EUT and monitoring equipment.	N/A	N/A		
2.0	Using the oscilloscope measurement function, measure the background network differential noise level. Record the result below. Vnoise = _____ mVrms	N/A	N/A		
3.0	Power on the EUT. Using the oscilloscope measurement function, measure the background network differential noise level. Record the result below. Vnoise = _____ mVrms	Vnoise < 60 mVrms			
4.0	Press the Service pushbutton on the EUT. Using the oscilloscope capture the transmit waveform. Using the measurement function, measure the amplitude of the transmit signal. Record the result below. Vsignal = _____ mVrms	Vsignal > 425 mVrms			

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
5.0	Power on the CDT. Power on the Traffic Generator to simulate all other nodes in the car. Allow network to initialize and behaviour of all traffic generator nodes to stabilize. Using the oscilloscope measurement function, measure the background network differential noise level. Ensure no portion of a transmit pulse is present in the measurement window. Record the result below. Vnoise = _____ mVrms	N/A	N/A		

8.2 EMI / EMC Compatibility – Common Mode Noise Immunity – Conducted Disturbance

PROCEDURE SCOPE	EMI / EMC compatibility – Common Mode Noise Immunity – Conducted Disturbance
PROCEDURE OBJECTIVE	Verify EUT is immune to conducted disturbances on the network cable shield.
REFERENCE DOCUMENTS	IEC 62236-3-2 Railway applications – Electromagnetic compatibility Section 8, Table 8. IEC 61000-4-6 Immunity to conducted disturbances induced by radio frequency fields
TEST REQUIREMENT	This test in conducted in accordance with IEC 61000-4-6, section 6.2.1 for screened cables. No functional impairment of the network operation at test level 3 (10 Vrms, 150 kHz to 80 MHz)
TEST FACILITY	This procedure shall take place in an electronic test and maintenance facility with ESD protection.
TEST PERSONNEL	Inspection and test technician with ESD suppression training.

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect the CDT, Traffic Generator, EUT and monitoring equipment. Set up the EUT in accordance with IEC 61000-4-6 and connect the disturbing signal generator to the network cable at the EUT. Place an isolator between the disturbance signal injection point at the EUT and the remainder of the devices on the network. Take a photograph of the test setup for inclusion in the report. Power on the Traffic Generator to simulate all nodes in Married pair configuration. Power on the CDT configured for Car A. Power on the EUT configured for Car A. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		
2.0	Start logging network traffic with Protocol Analyzer. Verify the CDT and EUT are working correctly. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends StatusBits with a 2 seconds transmission period (± 250 ms).			
		All EUT subscribed Parameter items are sent with the subscribed transmission period (± 250 ms).			
		The CDT reports no loss of communication with the EUT.			
		The EUT reports no loss of communication with the CDT.			
3.0	Set up CDT to report any change to the EUT StatusBits state or subscribed Parameter values.	N/A	N/A		

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
4.0	Turn on the signal generator and initiate the frequency scan. Using the oscilloscope to verify the magnitude and frequency of the disturbing signal. Monitor the network traffic for the duration of the scan. (approximately 40 minutes).	CDT reports no state or value changes during test.			
		The CDT reports no loss of communication with the EUT.			
		The EUT reports no loss of communication with the CDT.			

8.3 EMI / EMC Compatibility – Common Mode Noise Immunity – Ringing Wave

PROCEDURE SCOPE	EMI / EMC compatibility – Common Mode Noise Immunity – Ring Wave
PROCEDURE OBJECTIVE	Verify EUT is immune to conducted disturbances on the network cable shield.
REFERENCE DOCUMENTS	IEC 61000-4-6 Immunity to conducted disturbances induced by radio frequency fields
TEST REQUIREMENT	This test in conducted in accordance with IEC 61000-4-6, section 6.2.1 for screened cables but utilizes a repetitive, ringing impulse waveform to create the disturbance. The waveform comprises a continuous stream of high voltage impulses with a repetition rate of 19.5 kHz. The impulses are filtered to produce ringing waveforms with fundamental frequencies of 120 kHz, 240 kHz and 360 kHz and are applied for a one minute interval No functional impairment of the network operation with disturbing signal applied.
TEST FACILITY	This procedure shall take place in an electronic test and maintenance facility with ESD protection.
TEST PERSONNEL	Inspection and test technician with ESD suppression training.

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
1.0	Connect the CDT, Traffic Generator, EUT and monitoring equipment. Set up the EUT in accordance with IEC 61000-4-6 and connect the disturbing signal generator to the network cable at the EUT. Place an isolator between the disturbance signal injection point at the EUT and the remainder of the devices on the network. Take a photograph of the test setup for inclusion in the report. Power on the Traffic Generator to simulate all nodes in Married pair configuration. Power on the CDT configured for Car A. Power on the EUT configured for Car A. Allow network to initialize and behaviour of all traffic generator nodes to stabilize.	N/A	N/A		

SEQ #	ACTIONS, OBSERVATIONS, RESULTS	PASS / FAIL CRITERIA	PASS / FAIL	INITIAL	COM-MENT
2.0	Start logging network traffic with Protocol Analyzer. Verify the CDT and EUT are working correctly. Record the network traffic with Protocol Analyzer for up to two (2) minutes and save the log to file. Record the file name below. _____	EUT sends StatusBits with a 2 seconds transmission period (± 250 ms).			
		All EUT subscribed Parameter items are sent with the subscribed transmission period (± 250 ms).			
		The CDT reports no loss of communication with the EUT.			
		The EUT reports no loss of communication with the CDT.			
3.0	Set up CDT to report any change to the EUT StatusBits state or subscribed Parameter values.	N/A	N/A		
4.0	Turn on the signal generator and initiate the disturbing signal with a ringing frequency of 120 kHz. Using the oscilloscope verify the magnitude and frequency of the disturbing signal. Monitor the network traffic for one minute with the disturbing signal applied.	CDT reports no state or value changes during test.			
		The CDT reports no loss of communication with the EUT.			
		The EUT reports no loss of communication with the CDT.			
5.0	Turn on the signal generator and initiate the disturbing signal with a ringing frequency of 240 kHz. Using the oscilloscope verify the magnitude and frequency of the disturbing signal. Monitor the network traffic for one minute with the disturbing signal applied.	CDT reports no state or value changes during test.			
		The CDT reports no loss of communication with the EUT.			
		The EUT reports no loss of communication with the CDT.			
6.0	Turn on the signal generator and initiate the disturbing signal with a ringing frequency of 360 kHz. Using the oscilloscope verify the magnitude and frequency of the disturbing signal. Monitor the network traffic for one minute with the disturbing signal applied.	CDT reports no state or value changes during test.			
		The CDT reports no loss of communication with the EUT.			
		The EUT reports no loss of communication with the CDT.			

Appendix A: Comments

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#	SECTION #	SEQ #	COMMENT

Appendix B: Approval

After the completion of the interoperability test, all parties involved shall approve the results.

FULL NAME	AFFILIATION	ROLE	ATTENDED	COMMENTS	SIGNATURE	DATE